classic **Spring Boot file upload/download system design + implementation problem**

1. Understand the Requirements (Very Important)

✅You need to build:

POST /uploader

- Input:
  - fileGroup (String) = folder name
  - files (Multipart files) = contents inside folder
- Behavior:
  - Store files in DB
  - (fileGroup + fileName) = unique key , If same (fileGroup + fileName) → **replace existing**
  - Return **201**

GET /downloader?fileGroup=

- If:
  - ❌No files → **404**
  - ✅1 file → return file directly
  - ✅Multiple files → return **ZIP**

 Validations:

- File size > 100KB → ❌reject (500 error)
- Use **config**, not manual check

 2. High-Level Design

Controller → Service → Repository → DB (H2)

Google Drive:
 certificates
- degree.pdf
- marksheet.pdf

 invoices
- jan.pdf
- feb.pdf

What does “upload a file” mean here?

You are sending a request like this to your backend:

fileGroup = "certificates"
files = [degree.pdf]
 The backend receives:

- file name → degree.pdf
- file content → bytes

What Happens Internally?

When you click **Send**:

Postman sends:

Content-Type: multipart/form-data
With data like:

fileGroup = certificates
files = degree.pdf (binary)
Spring Boot receives:
@RequestParam String fileGroup
@RequestParam List<MultipartFile> files
 Automatically converts:

- PDF → MultipartFile
- File content → byte[]

✅Expected Result

- Status: **201 Created**
- File stored in DB

** Verify It Worked**

Open browser: http://localhost:8080/h2-console

Run: SELECT * FROM XFILE;

 You should see:

- fileGroup = certificates
- fileName = degree.pdf

What your API is actually doing

Case 1: Single file

Content-Disposition: attachment; filename=degree.pdf
 Browser/Postman should treat it as a file

Case 2: Multiple files

Content-Disposition: attachment; filename=certificates.zip
 Returns ZIP bytes